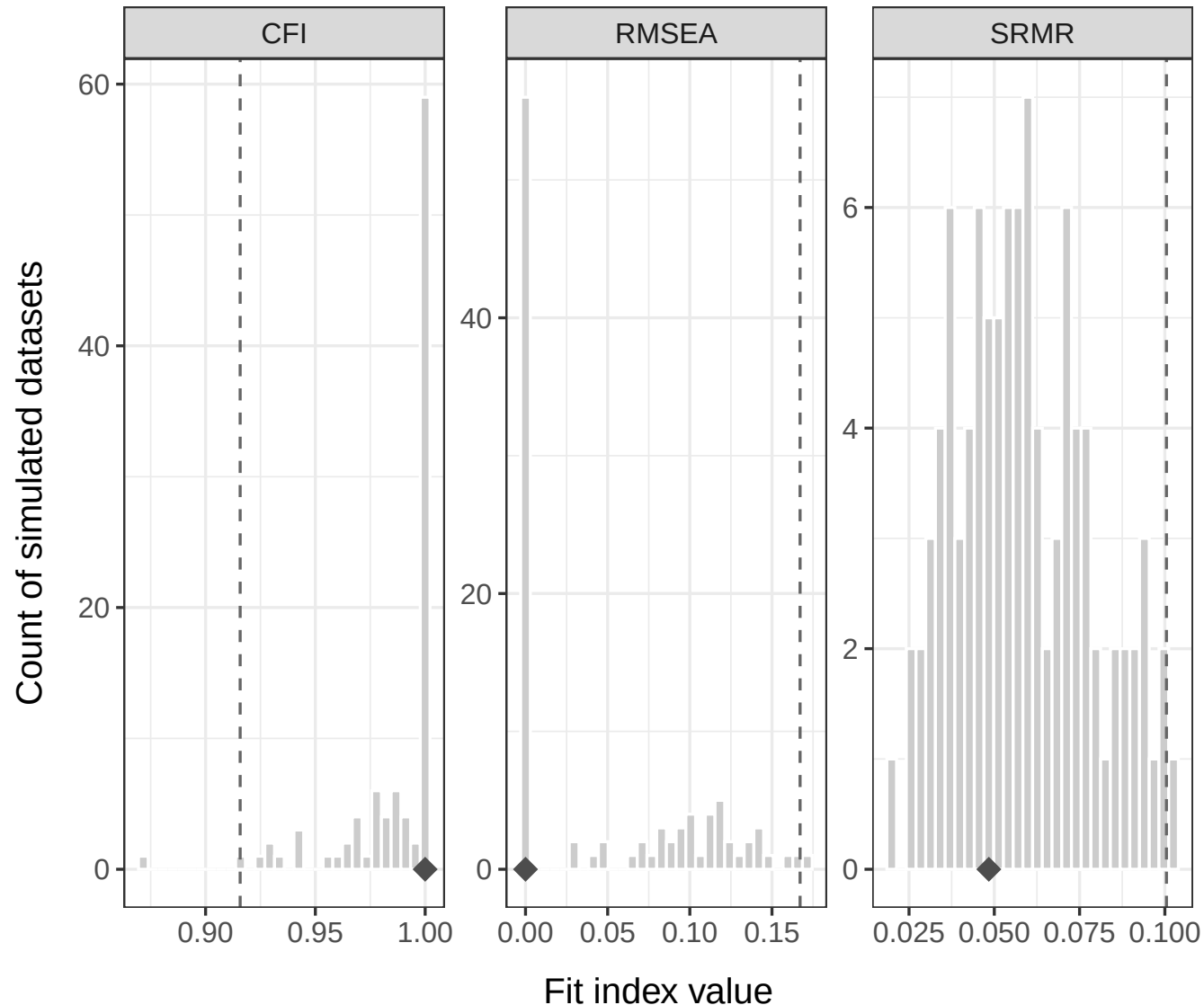
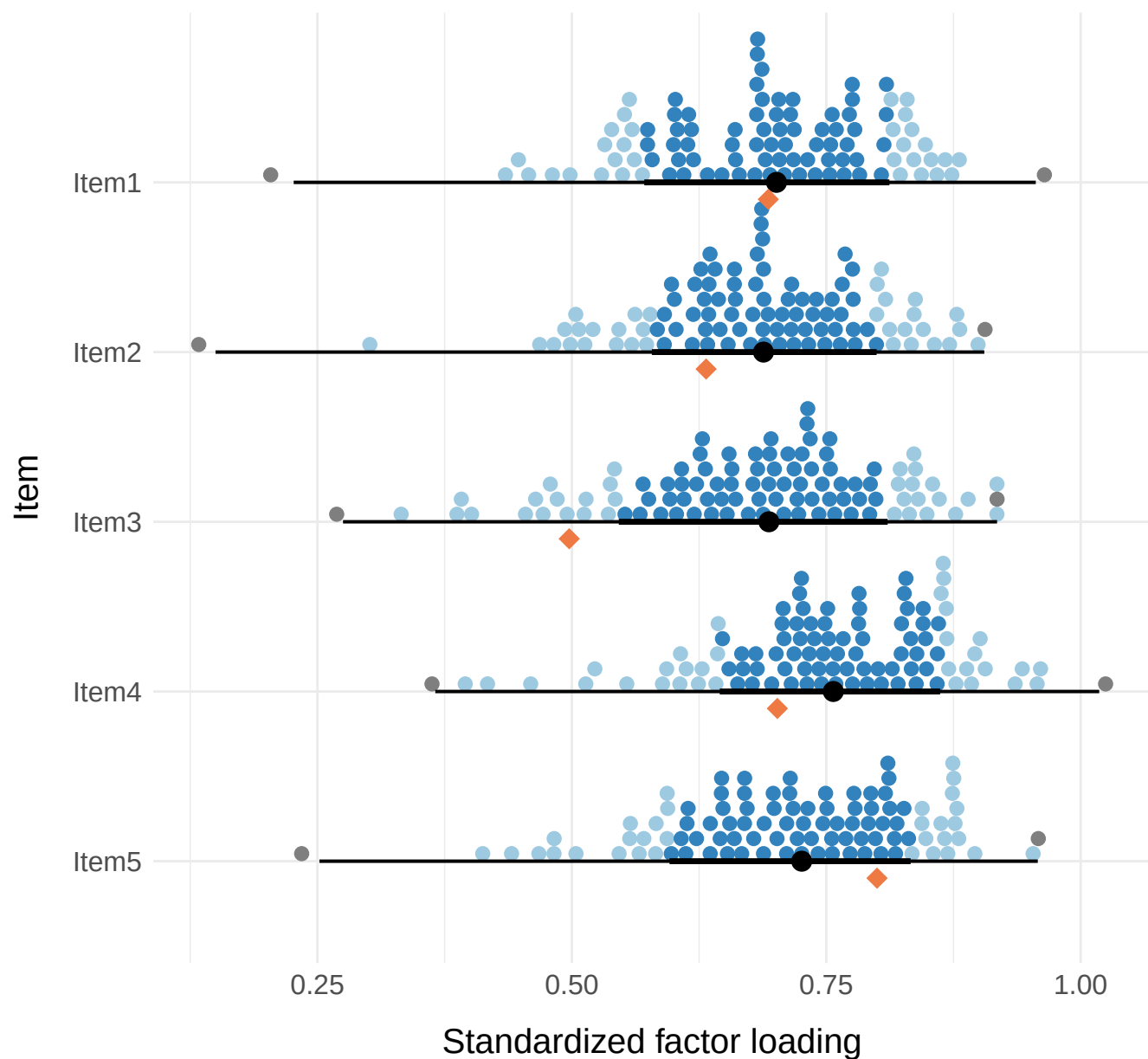


Observed CFA fit vs simulated null distribution

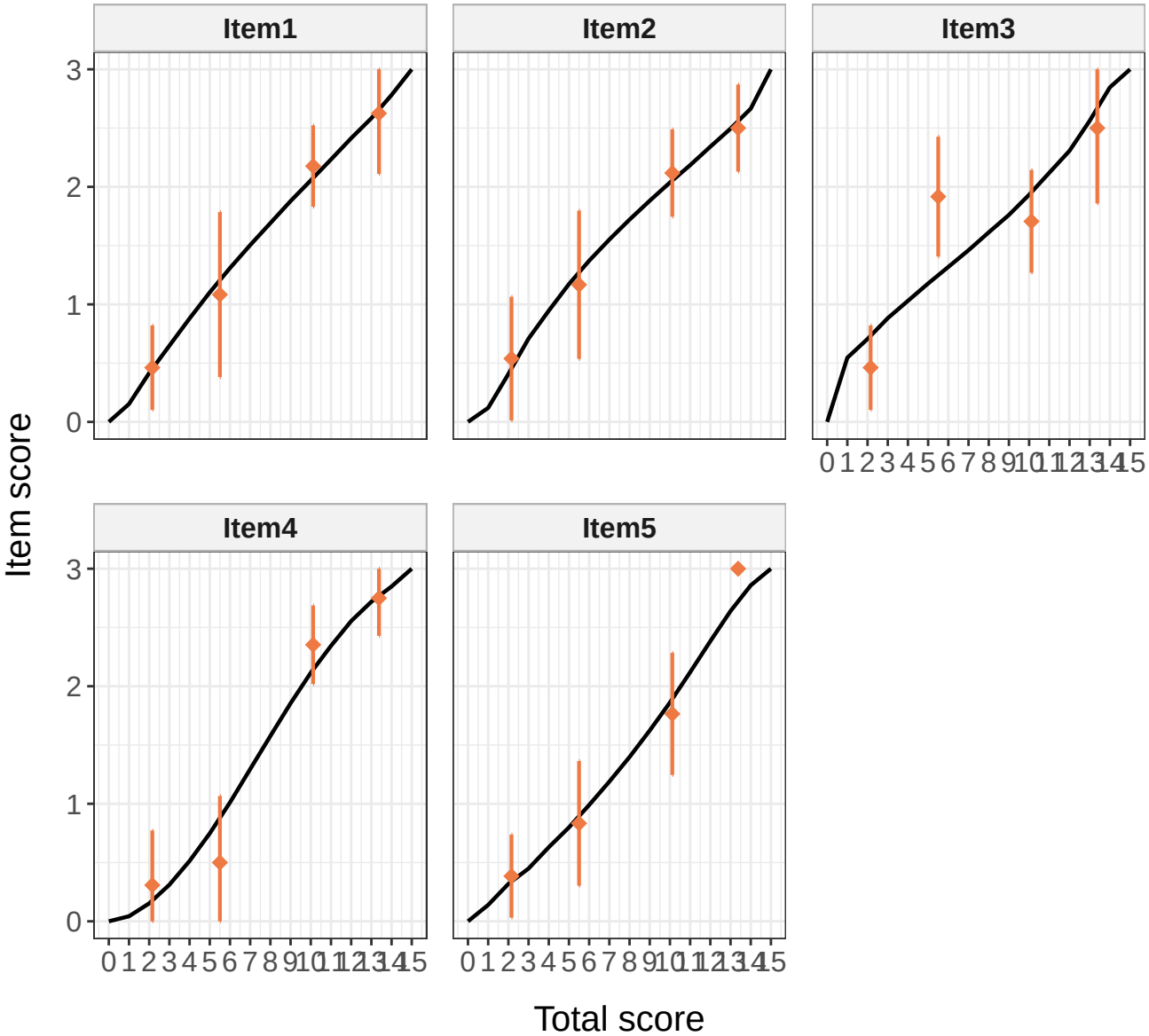


Note. Histograms: 99 datasets simulated under PCM unidimensionality at $n = 50$ respondents per dataset, refitted with `lavaan::cfa(ordered = TRUE, estimator = ...WLSMV...)`. Diamond: observed value (red = flagged at the 99th percentile in the unfavourable direction). Dashed line: cutoff (CFI: 1th pct; RMSEA / SRMR: 99th pct).



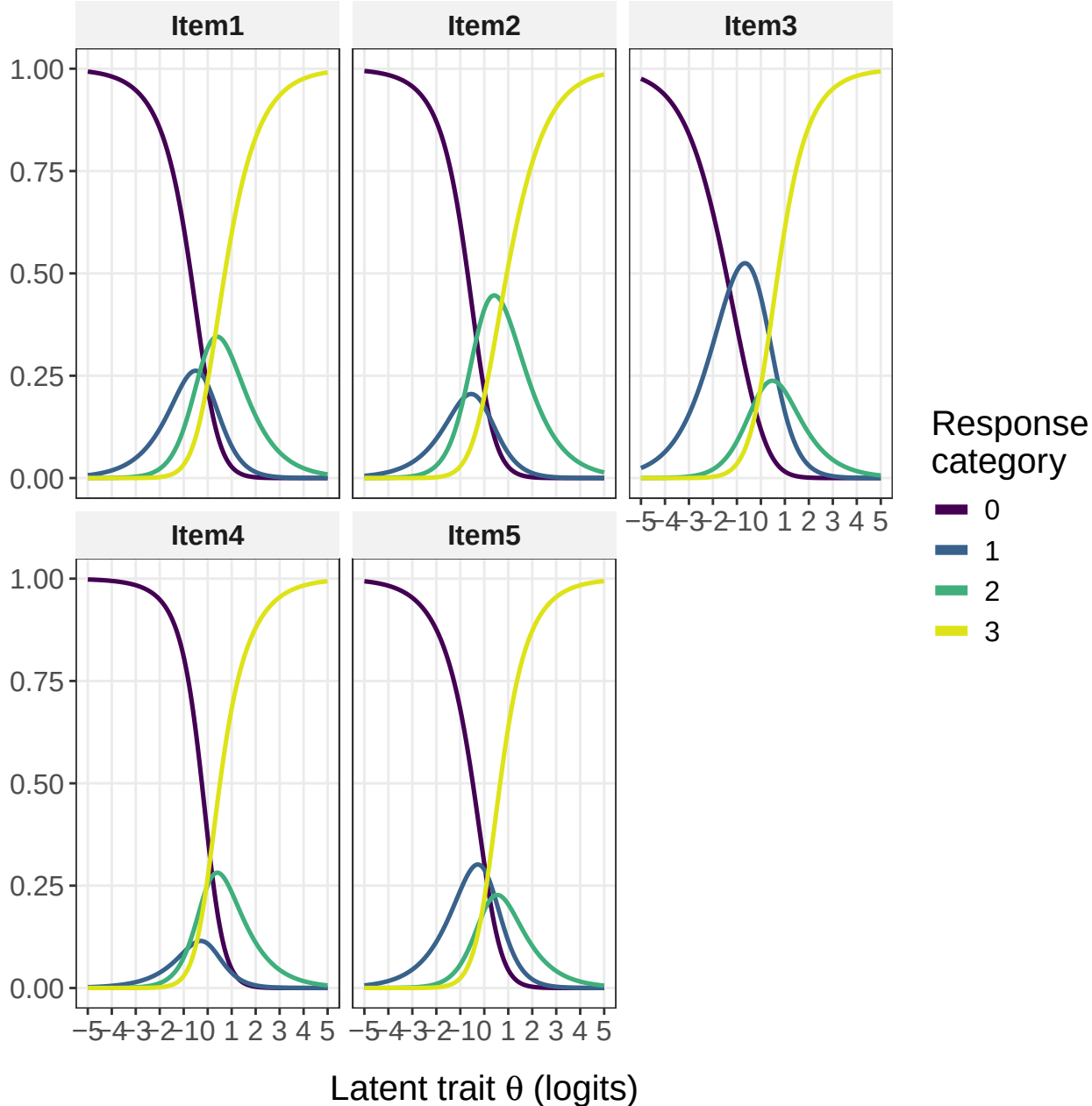
Note. Standardized factor loadings from 99 datasets simulated under PCM unidimensionality at $n = 50$ respondents per dataset, refitted with `lavaan::cfa(ordered = TRUE, estimator = ...WLSMV...)`. Diamonds: observed loading (red = outside the two-sided 99% expected range). Black dots: simulated median.

Conditional Item Characteristic Curves

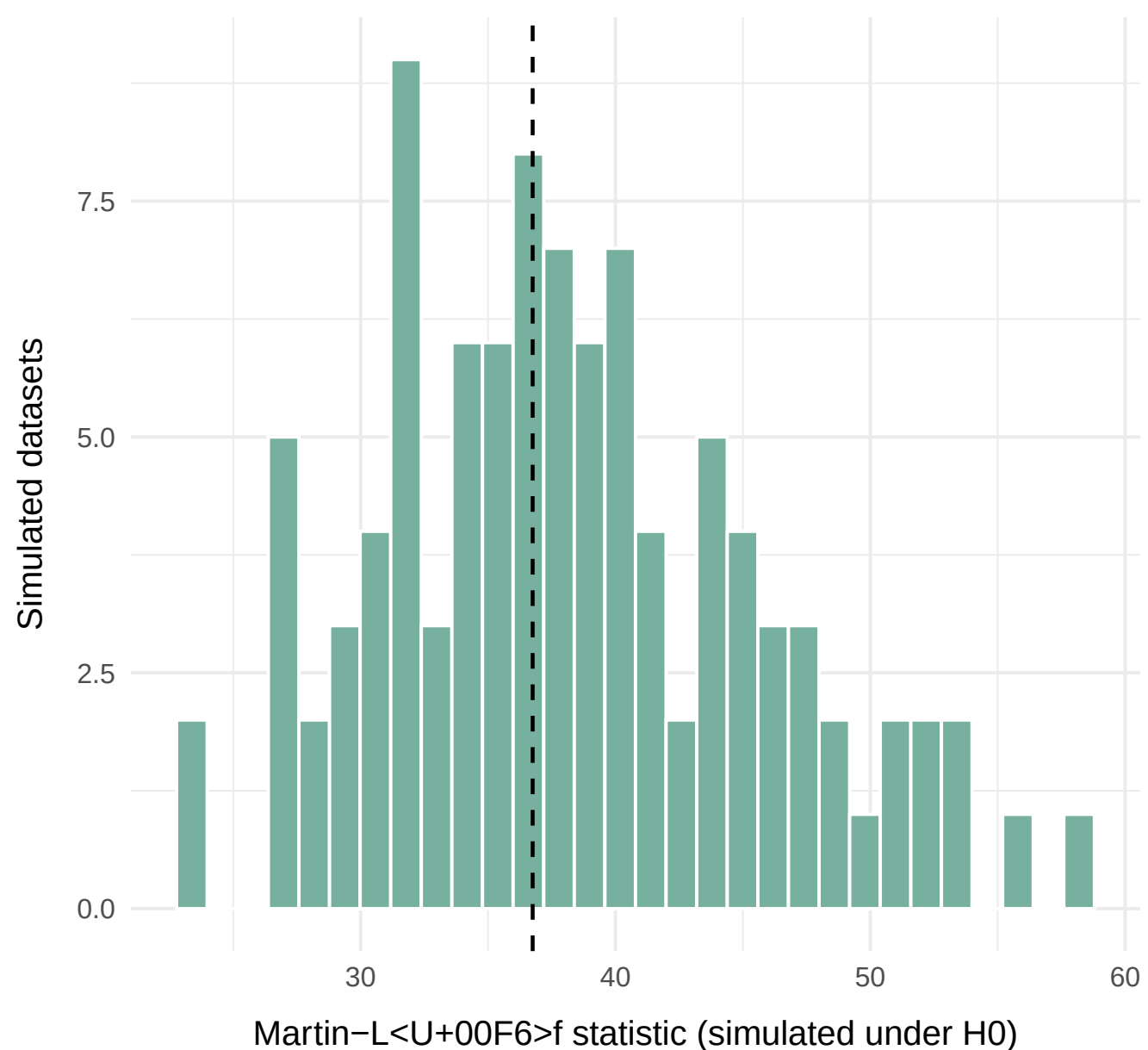


Note. Black line: model-expected conditional item score. Diamonds: observed mean item score per total-score group with 95% confidence-interval error bars.

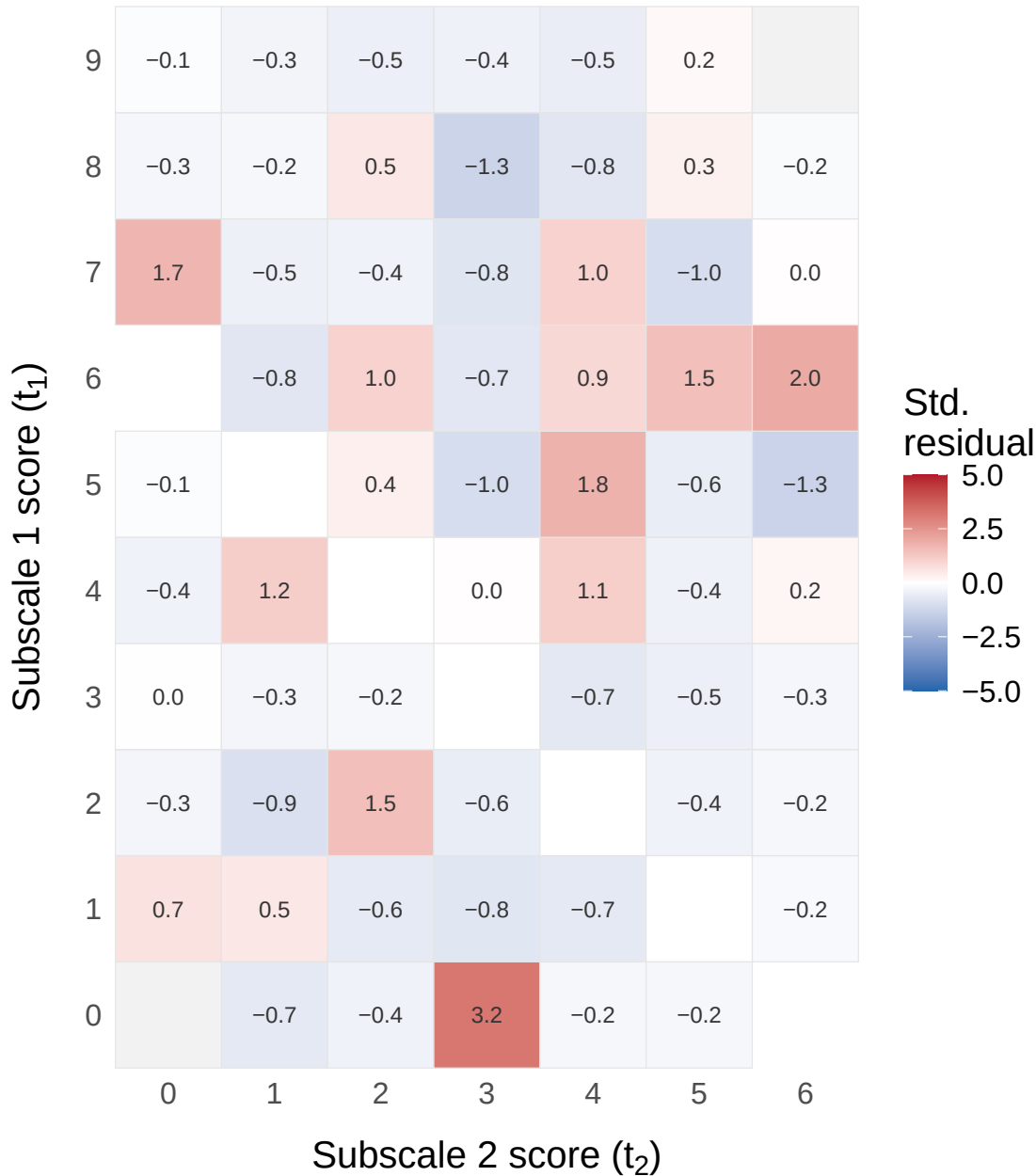
Category probability



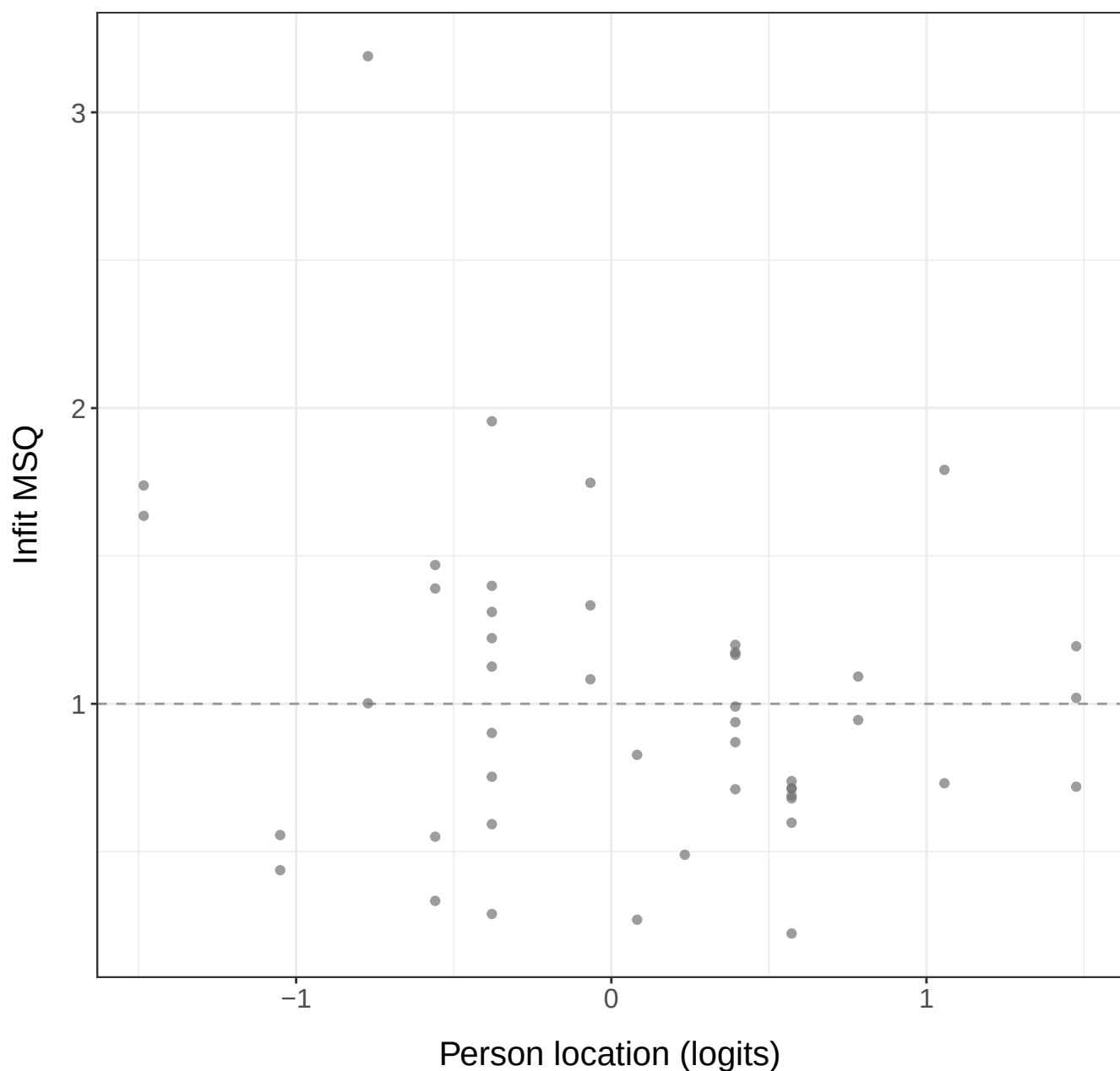
Note. n = 50 respondents.



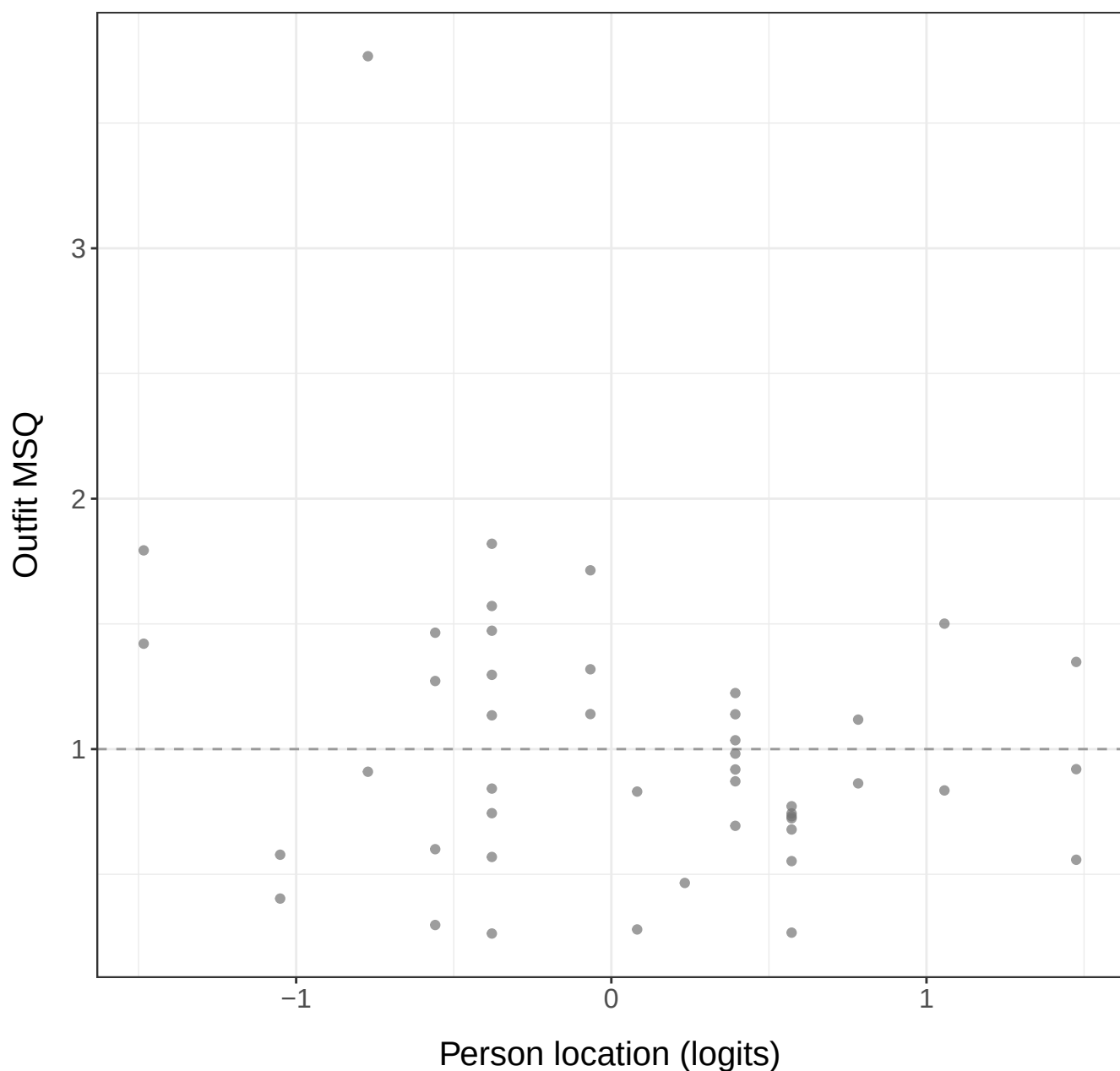
Note. Martin-L $\langle U+00F6 \rangle f$ statistics from 100 datasets simulated under the unidimensional null ($n = 50$ per dataset). Dashed line: observed $T = 36.75$; Monte Carlo $p = 0.554$. More iterations are generally recommended for publication-ready results.



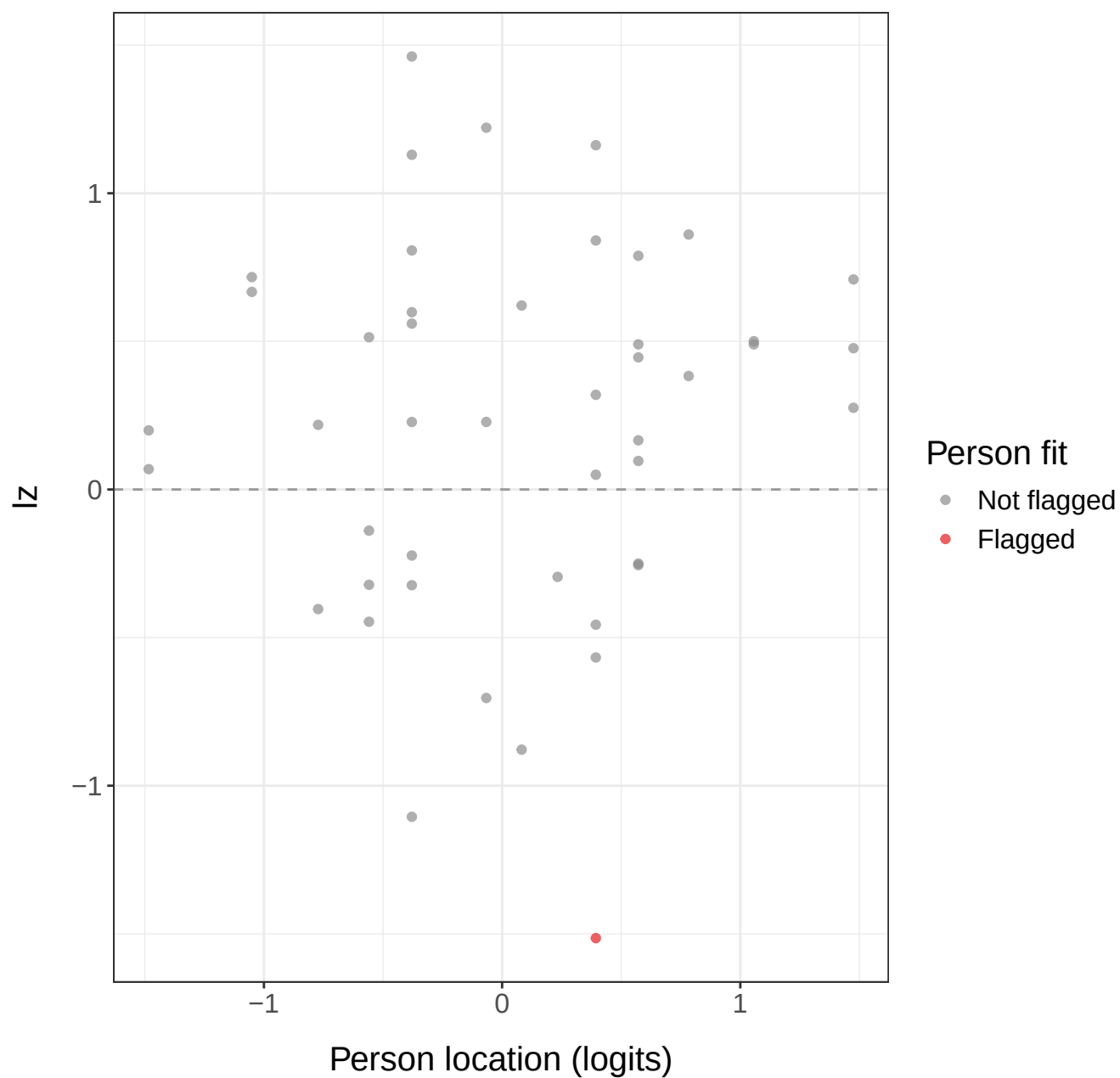
Note. Standardised residuals (Christensen et al. 2002, eq. 13). $n = 50$ respondents.
Cells with $|\text{residual}| > 2$ indicate potential dimensionality issues.



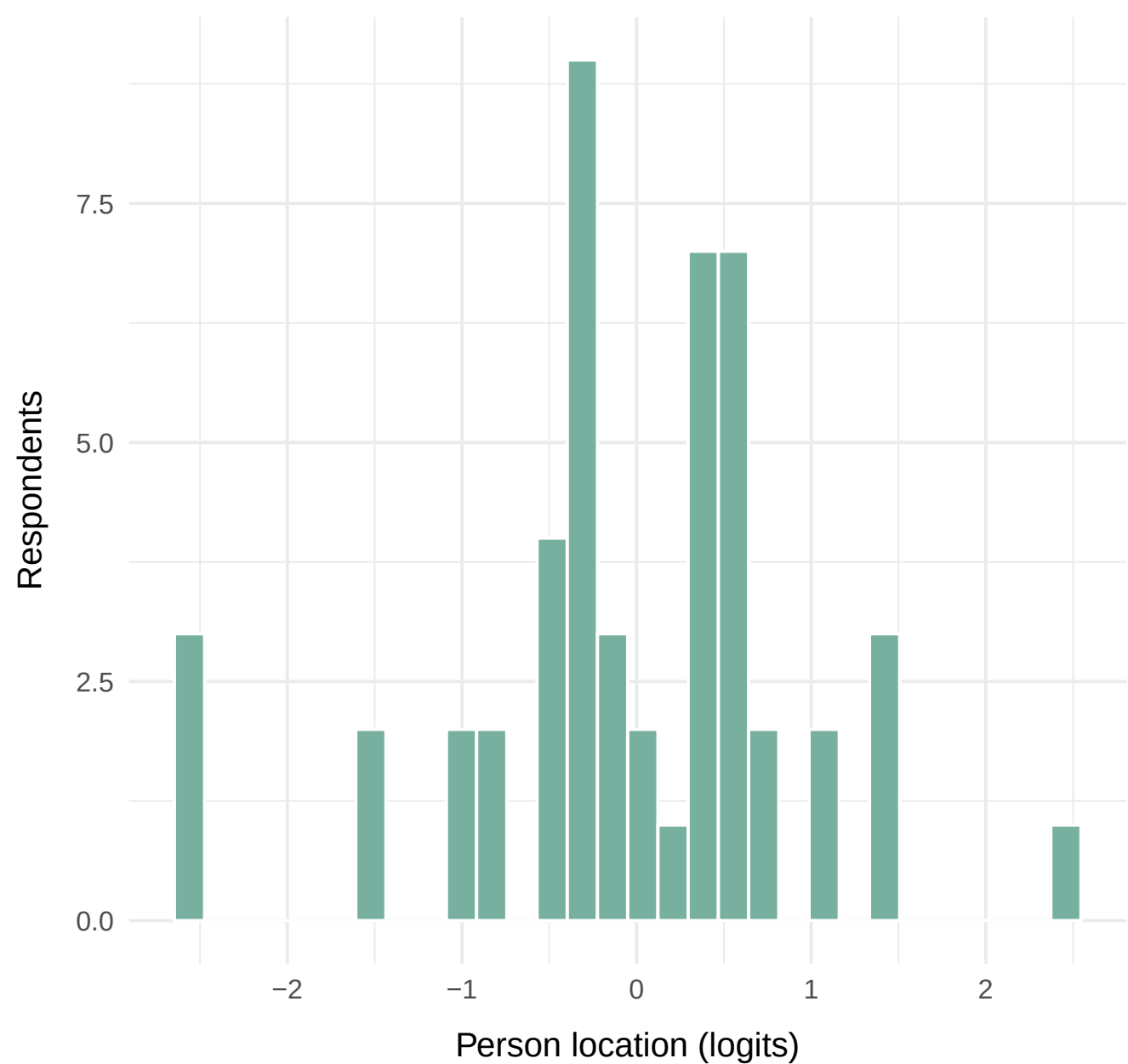
Note. n = 46 of 50 respondents assessed (extreme scores excluded); 0 (0%) flagged: 0 underfit (MSQ > 1), 0 overfit (MSQ < 1) ($p < 0.05$).



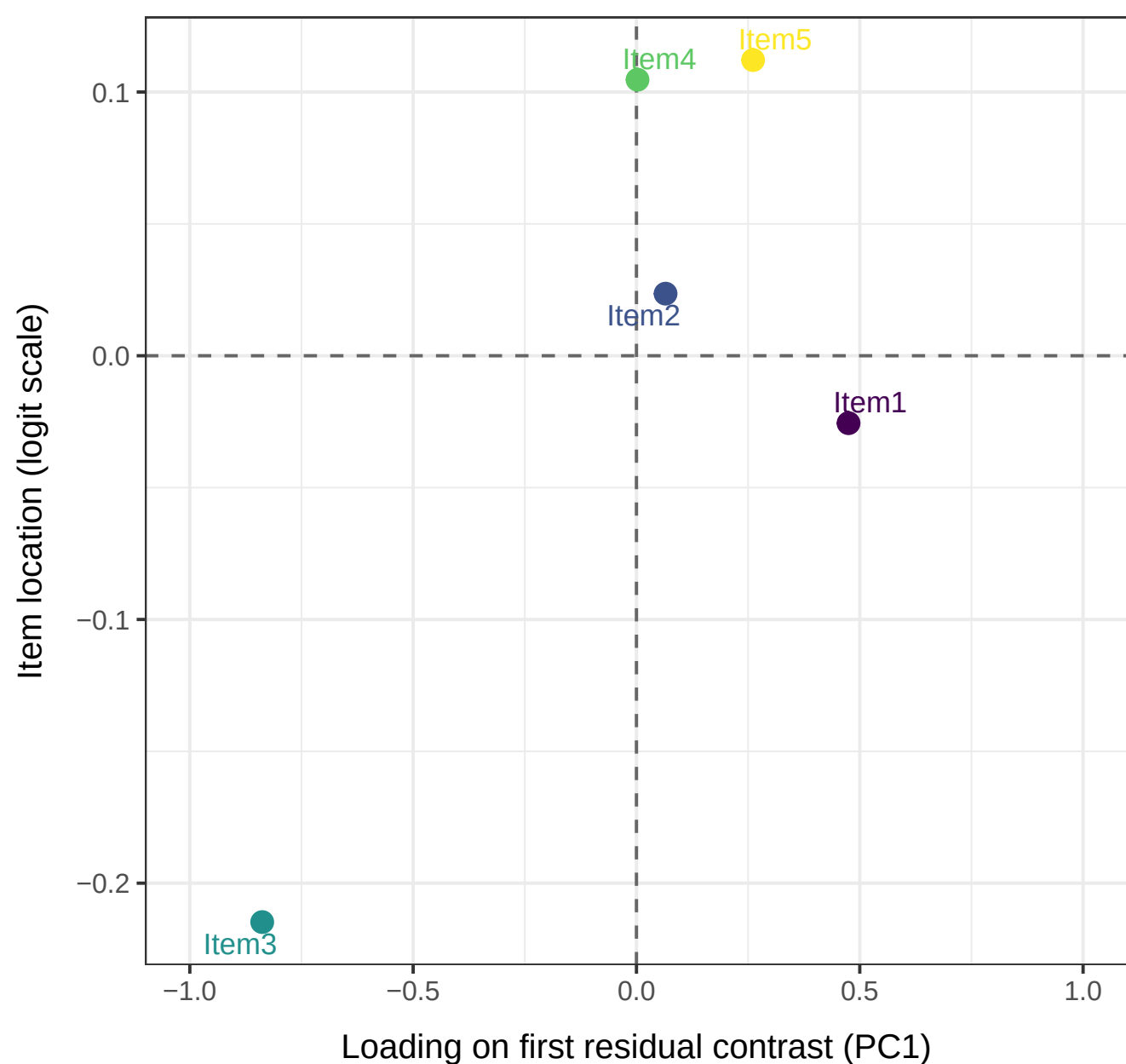
Note. $n = 46$ of 50 respondents assessed (extreme scores excluded); 0 (0%) flagged: 0 underfit ($MSQ > 1$), 0 overfit ($MSQ < 1$) ($p < 0.05$).



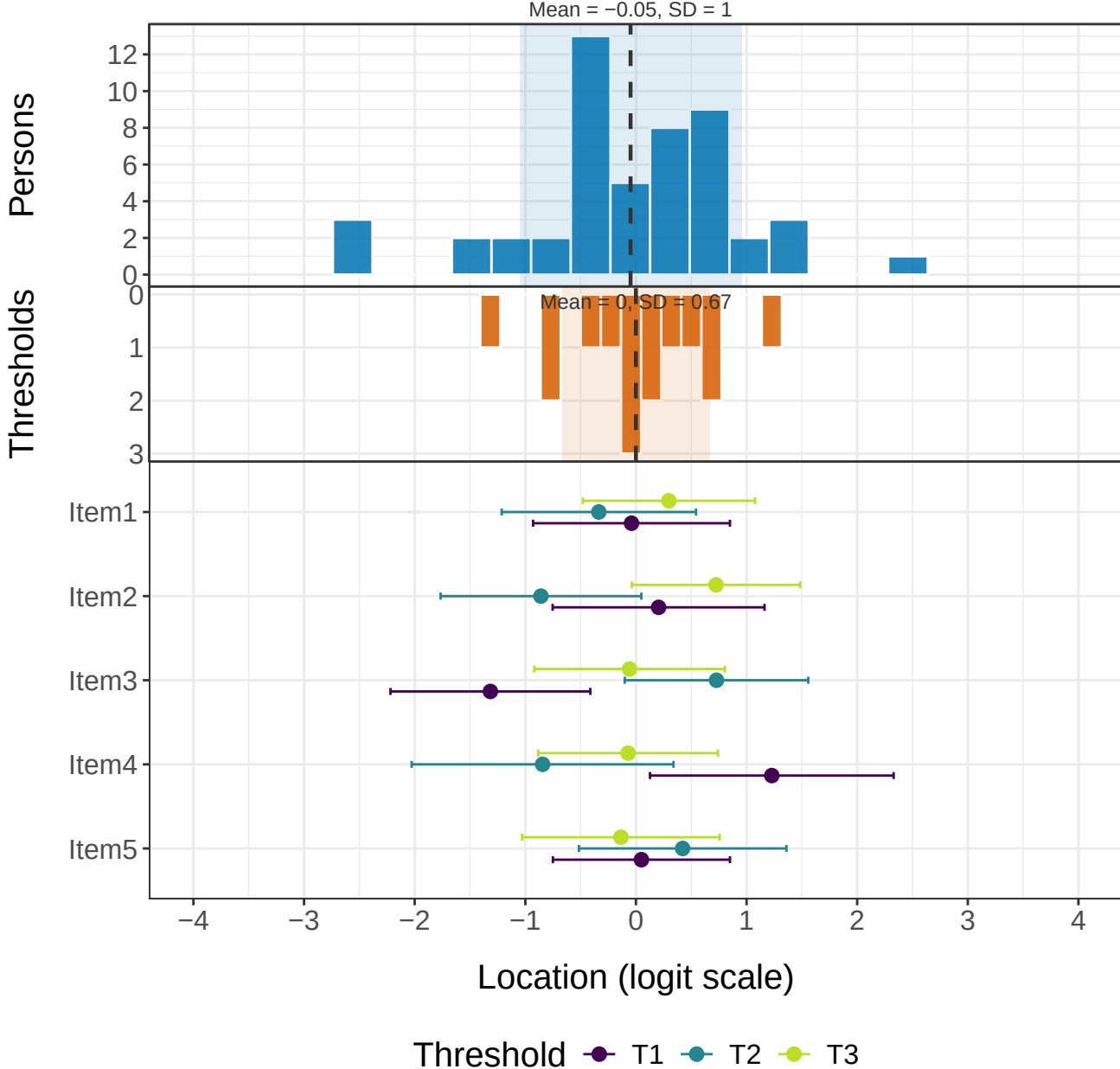
Note. n = 46 of 50 respondents assessed (extreme scores excluded); 1 (2.2%) flagged as misfitting ($p < 0.05$).



Note. WLE estimates for $n = 50$ respondents. Minimum score: $n = 3$ (6.0%); maximum score: $n = 1$ (2.0%).



Note. Total observed variance: 41.6% explained by measures, 58.4% unexplained (basis for PCA; n = 50 respondents, WLE).



Note. Person location mean: -0.05 (SD 1). Item threshold location mean: 0 (SD 0.67). n = 50 respondents. Error bars show 95% confidence intervals.